

Task:

Identify which player within a Pokemon VGC match wins based on the pokemon that each player leads off with in a match.

Unit of analysis:

The unit of analysis is the percentage chance of player 1 winning and compare it to the winner of the battle

Features:

Each battle has four positions and 2 pokemon belong to each player

P1a is player 1's first pokemon, P1b is player 1 second pokemon, P2a is player 2's first pokemon, and P2b is player 2's second pokemon.

Understanding this, the list of features is:

p1a, p1a item, p1a ability, p1a moves, p1a nature,
p1b, p1b item, p1b ability, p1b moves, p1b nature,
p2a, p2a item, p2a ability, p2a moves, p2a nature,
p2b, p2b item, p2b ability, p2b moves, p2b nature

Train/Validation/Test strategy:

Our strategy will go off a 80% training, 10% validation, 10% test split and will train off of "logs_gen9championsvgc2026regma.json" and "logs_gen9championsvgc2026regmb.json"

Feasibility:

In terms of the amount of quality data, we have a big enough source so we can pull more data if we need more. The main problem with a VGC battle predictor is the sheer dimensionality due to the many variables a single pokemon can have. By reducing it to just the lead pokemon we reduce the dimensionality enough where it's manageable to predict who wins.

Technical / Compute requirements

After processing the data, the dataset we are planning to train with is reduced to be less than 10MB (8k~ entries), therefore making it easier to compute on a small end laptop.

Source Ethics and Considerations:

Source does not collect data from human participants and does not sensitive personal data.